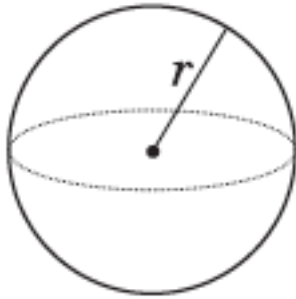


8.6/8.7 Volume and Surface Area of a Sphere

MPM1D

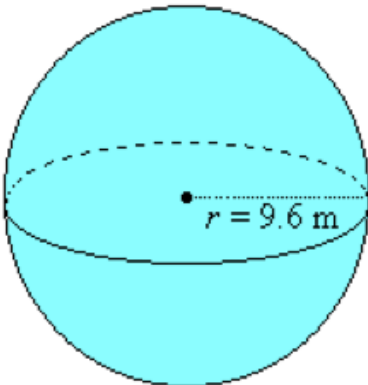
Part A: Volume of a Sphere

Sphere



$$Volume_{sphere} = \frac{4 \pi r^3}{3}$$

Example 1: Find the volume of the following sphere



Volume_{sphere} = _____

Example 2: Find the volume of a sphere with a radius of 5 km

Example 3: A city water storage tank is in the shape of a sphere. The radius of the sphere is 12.5 meters.

a) Find the volume of the storage tank in cubic meters



b) Find the volume of the storage tank in Litres

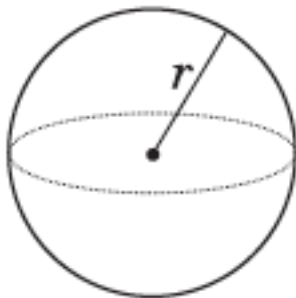
$$1\text{L} = 1000 \text{ cm}^3$$

and

$$1 \text{ m}^3 = 1000 \text{ L}$$

Part B: Surface Area of a Sphere

Sphere



$$\text{Surface Area}_{\text{sphere}} = 4\pi r^2$$

Example 4: The dimensions of an adult human eyeball are reasonably constant, varying only by a millimeter or two. The average diameter is about 2.5 cm. Calculate the surface area of the human eyeball, to the nearest tenth of a square centimeter.



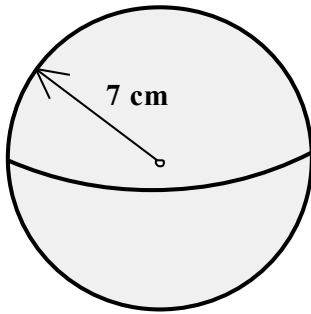
Example 5: Determine the radius of a baseball that has a surface area of 215 cm^2 . Round your answer to the nearest tenth of a centimeter.



Part C: Applications

Example 6: Determine the **volume and surface area** of the given spheres.

a)



b)

